Forks in Blockchain

Defination

- A fork is a code modification of a blockchain that is similar to the original blockchain.
- There are different types of forking
- A hard fork : is a radical change in a cryptocurrency protocol that is incompatible with the previous blockchain versions.
- A soft fork : is a change in a cryptocurrency protocol that keeps it backward compatible.

Fork...

- In s/w development,there is a central authority that is running the entire show.
- A group or individual is creting the whole syatem,
- in case of maintainance, the same group is taking charge of it.
- But blockchain does not follow the life cycle of Web 2.0.

Blockchain

- Once a product of of blockchain is developed,tested and released, it doesn’t remain in the hand s of the developer ,even the creator.

- So how do blockchain get upgraded?

Fork

- Visualize blockchain as a chain of blocks stacked over one another.
- It is meant for storing data.
- Now there are rules theat govern for creation of blocks,related to size, no of blocks,transactionsetc,

Blockchain

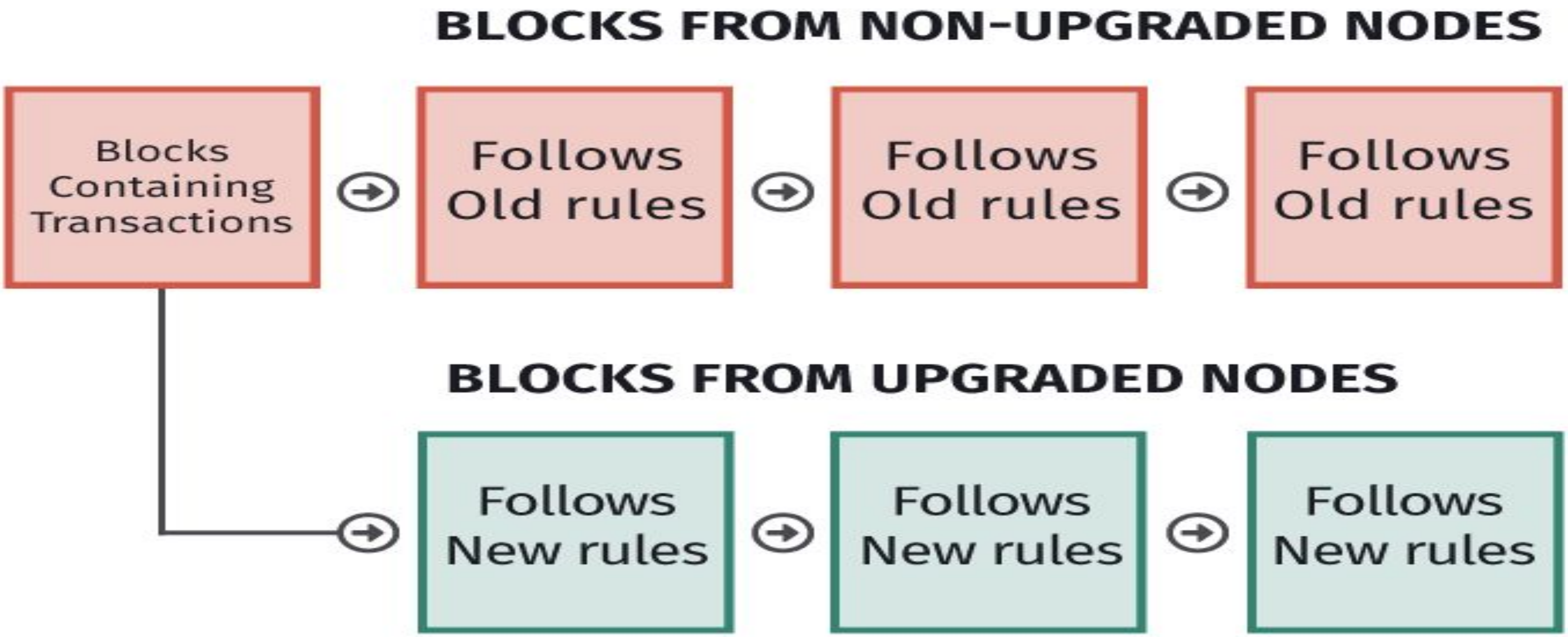
- Every individual has a copy of blockchain , onre may edit the rules any time.

- This would create a new blockchain

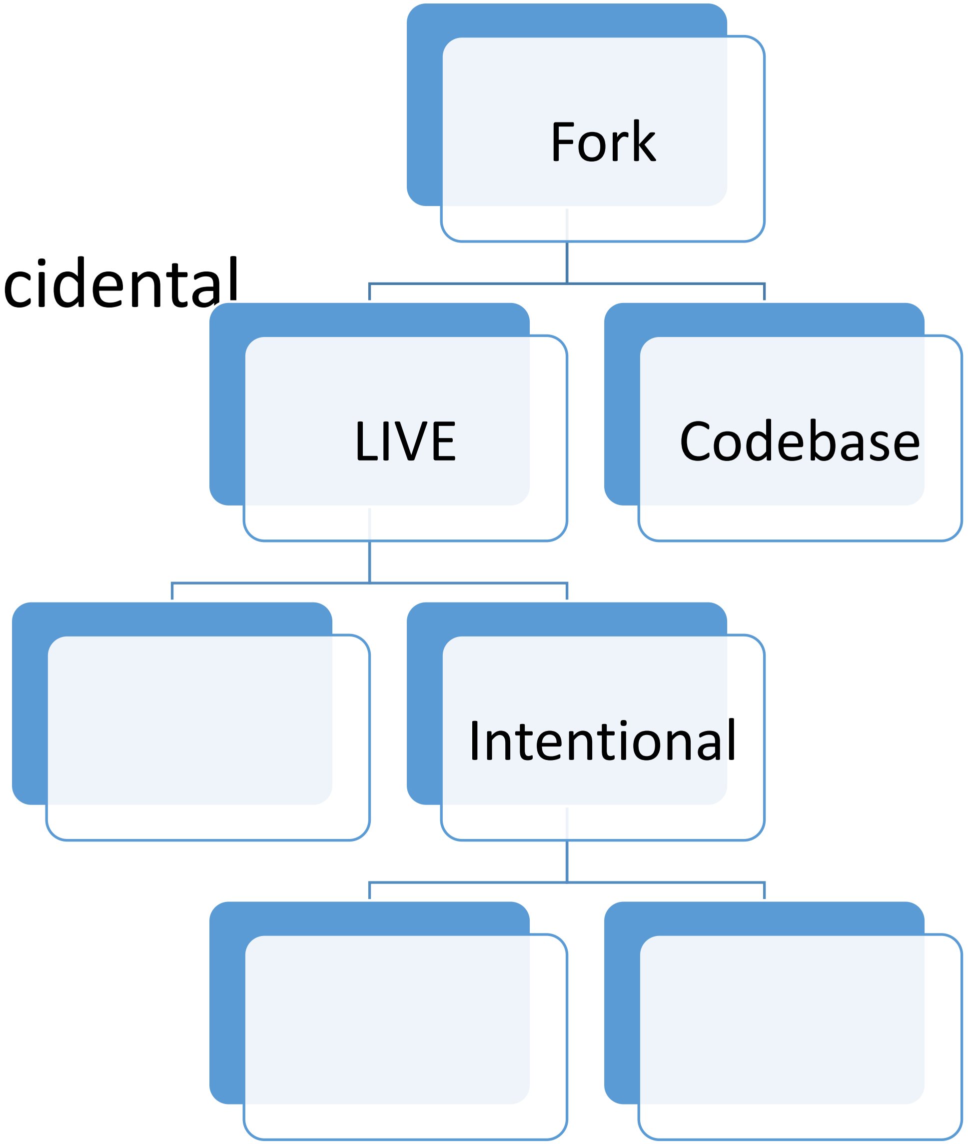
Types of fork

- Soft
- Hard

Fork


BLOCKS FROM NON-UPGRADED NODES
Blocks Containing Transactions
Follows Old rules
Follows Old rules
Follows Old rules
BLOCKS FROM UPGRADED NODES
Follows New rules
Follows New rules
Follows New rules

- Intentional and Accidental
- Intentional ->
- Soft and Hard



Hard

- When the blockchain protocol is altered in a non backwards-compatible way.
- When there is a change in the software that runs on the full nodes to function as a network participant, the change is such that the new blocks mined on the basis of new rules (in the Blockchain protocol) are not considered valid by the old version of the software.
- When hard forks occur, new currency come into existence (with valid original currency) like in the case of Ethereum (original : Ethereum, new : Ethereum Classic) and Bitcoin (original : Bitcoin, new : Bitcoin cash).

Soft

- When the blockchain protocol is altered in a backwards-compatible way. In soft fork you tend to add new rules such that they do not clash with the old rules.
- there is no connection between the old rules and new rules.
- Rules in soft fork are tightened.
- When there is a change in the software that runs on the nodes (better called as ‘full nodes’) to function as a network participant, the change is such that the new blocks mined on the basis of new rules (in the Blockchain protocol) are also considered valid by the old version of the software.
- This feature is also called as backwards-compatibility.

- The Bitcoin network's SegWit update added a new class of addresses (Bech32). However, this didn't invalidate the existing P2SH addresses. A full node with a P2SH type address could do a valid transaction with a node of Bech32 type address.